

E - Endless Holidays

分值: 450 分

Problem Statement

The Kingdom of AtCoder has N cities, called city $1, 2, \dots, N$. There are M bidirectional roads connecting pairs of cities, where the i -th road connects cities U_i and V_i . Any pair of cities can be reached from each other by traversing some roads.

AtCoder 王国有 N 座城市, 编号为 $1, 2, \dots, N$ 。共有 M 条双向道路连接城市对, 其中第 i 条道路连接城市 U_i 和 V_i 。任意两座城市都可以通过若干条道路互相到达。

In the Kingdom of AtCoder, a week consists of W days. A week proceeds through days $1, 2, \dots, W$, and the day after day W is day 1.

在 AtCoder 王国, 一周包含 W 天, 按顺序为 $1, 2, \dots, W$, 且第 W 天的下一天是第 1 天。

Each city has certain days of the week that are holidays. The holiday information for city i is given as a string S_i of length W : if the j -th character of S_i is o, day j is a holiday; if it is x, day j is a weekday.

每座城市在一周中有固定的休息日。城市 i 的休息日信息由一个长度为 W 的字符串 S_i 给出: 若 S_i 的第 j 个字符为 o, 则第 j 天是休息日; 若为 x, 则第 j 天是工作日。

Takahashi chooses a city he likes and visits it at noon on day 1. Each night thereafter, he repeatedly chooses to either stay in his current city or move to a city directly connected by a road. Output Yes if it is possible for him to keep moving so that the city he is in at noon every day is a holiday, and No otherwise.

高桥选择一个他喜欢的城市, 在第 1 天的中午抵达该城市。之后每天晚上, 他可以选择留在当前城市, 或者移动到与当前城市直接有道路相连的城市。如果存在一种移动方式, 使得他每天中午所在的城市当天都是休息日, 则输出 Yes, 否则输出 No。

T test cases are given; solve each of them.

共有 T 组测试数据, 你需要对每组数据分别求解。

Constraints

- $1 \leq T \leq 10^5$
 - $1 \leq N \leq 10^5$
 - $N - 1 \leq M \leq 10^5$
 - $1 \leq U_i < V_i \leq N$
 - Any pair of cities can be reached from each other by traversing some roads.
 - $2 \leq W \leq 10$
 - T, N, M, U_i, V_i, W are integers.
 - S_i is a string of length W consisting of o, x.
 - The sum of N over all test cases is at most 10^5 .
 - The sum of M over all test cases is at most 10^5 .
 - $1 \leq T \leq 10^5$
 - $1 \leq N \leq 10^5$
 - $N - 1 \leq M \leq 10^5$
 - $1 \leq U_i < V_i \leq N$
 - 任意两座城市都可以通过若干条道路互相到达。
 - $2 \leq W \leq 10$
 - T, N, M, U_i, V_i, W 是整数。
 - S_i 是长度为 W 的字符串，仅由 o 和 x 组成。
 - 所有测试数据的 N 之和不超过 10^5 。
 - 所有测试数据的 M 之和不超过 10^5 。
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Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

T
 case₁
 case₂

$\vdots$
 case_T

Here, case_i denotes the input for the i -th test case. Each test case is given in the following format:

其中 case_i 表示第 i 组测试数据的输入。每组测试数据按照以下格式给出：

$N\ M$
 $U_1\ V_1$
 $U_2\ V_2$
 $\vdots$
 $U_M\ V_M$
 W
 S_1
 S_2
 $\vdots$
 S_N

Output

Output T lines. The i -th line should contain the answer for the i -th test case.

输出 T 行，第 i 行应包含第 i 组测试数据的答案。

Sample Input 1

```
3
4 4
1 2
1 4
2 4
2 3
3
xxo
xox
oxo
oxx
1 0
4
```

```
0000
5 5
1 4
2 3
4 5
3 4
2 5
7
OXXXXXX
XXOXXO
XXXOXOX
XOXXOXX
OXXOXOX
```

Sample Output 1

```
Yes
Yes
No
```

For the first test case, for example, the condition can be satisfied by repeatedly moving along cities $4 \rightarrow 2 \rightarrow 1 \rightarrow 4 \rightarrow 2 \rightarrow 1 \rightarrow \dots$. Alternatively, the condition can also be satisfied by repeatedly moving along cities $3 \rightarrow 2 \rightarrow 3 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow \dots$.

对于第一组测试数据，例如，按顺序反复移动经过城市 $4 \rightarrow 2 \rightarrow 1 \rightarrow 4 \rightarrow 2 \rightarrow 1 \rightarrow \dots$ 即可满足条件。另外，按顺序反复移动经过城市 $3 \rightarrow 2 \rightarrow 3 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow \dots$ 也可以满足条件。

For the second test case, the condition can be satisfied by staying in city 1 indefinitely.

对于第二组测试数据，无限期停留在城市 1 即可满足条件。

For the third test case, it is impossible to keep moving while satisfying the condition.

对于第三组测试数据，不存在满足条件的持续移动方式。